

BioSeek: A Design Generation Framework of Biosignal Processors with Large-Language Models for Edge Healthcare Applications

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Abstract—Deep neural network (DNN)-based methodologies have shown impressive performance and robustness in the detection of abnormalities and decoding of multi-modal biosignals. While the use of DNNs provides promising classification and decoding capabilities, it also introduces significant design and cost challenges for the implementation of biomedical System on Chips (SoC). To address the increasing demand for advanced and efficient DNN-based healthcare solutions at the edge, we propose BioSeek, an agile design generation framework enhanced by cutting-edge large-language models (LLM). BioSeek offers a comprehensive solution to the design challenges associated with biosignal processors. The effectiveness of BioSeek is evaluated through the design generation of both application-specific and versatile biosignal processors, demonstrating performance that is competitive with existing solutions.

Index Terms—Biosignal Processing, System-on-Chip, Agile Design, Large Lanuage Model

I. INTRODUCTION

Interpreting biomedical signals is crucial for healthcare applications involving implantable and wearable devices, particularly in real-time disease diagnosis and brain-computer-interface (BCI) [1]. Research has highlighted the effectiveness of capturing multi-modal biosignals through front-end sensors [2]. Recently, systems utilizing artificial intelligence (AI) have emerged, enabling accurate and scalable classification and decoding of biosignals. Specialized deep neural networks (DNNs), including convolutional neural networks (CNNs) and spiking neural networks (SNNs), have been developed and implemented in edge healthcare systems [3] [4] [5] [6] [7]. To create more compact healthcare solutions at the edge, recent studies indicate a growing trend toward integrating front-end sensors and DNN processors as system-on-chip (SoC) solutions [8] [9] [10] [11] for higher energy/area efficiency and response speed.

Figure 1 illustrates the standard architecture of an AI-based biomedical SoC. Human-originated biosignals undergo amplification by a low-noise amplifier (LNA) and digitization by an analog-to-digital converter (ADC) within the multi-channel analog front-end (AFE) module [2]. The resulting temporal data sequences are stored and forwarded to the back-end neural signal processing unit, where DNNs are utilized

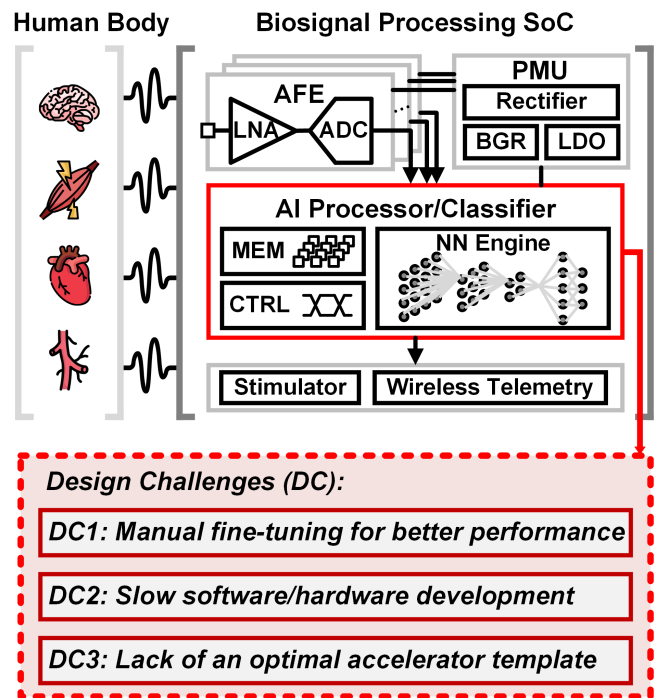


Fig. 1. Standard architecture of an AI-based biomedical SoC with AFE, PMU, AI processor, and stimulator/wireless telemetry.

for signal classification or decoding [3]. Subsequently, the classification/decoding outcomes are transmitted to a stimulator or wireless telemetry for closed-loop intervention [12]. Design optimizations of such SoC are targeted at pushing the pareto-frontier of the trade-off between classification/decoding performance and on-chip power/area/latency consumptions [4] [5] [6] [7] [9] [10] [11] [12].

Utilization of DNN brings promising classification/decoding performance as well as significant design/cost burdens for the biomedical SoC. From a design perspective, the main challenge is the complexity of designing and optimizing the model/processor architecture to efficiently support the computational demands of neural networks. Furthermore, from a

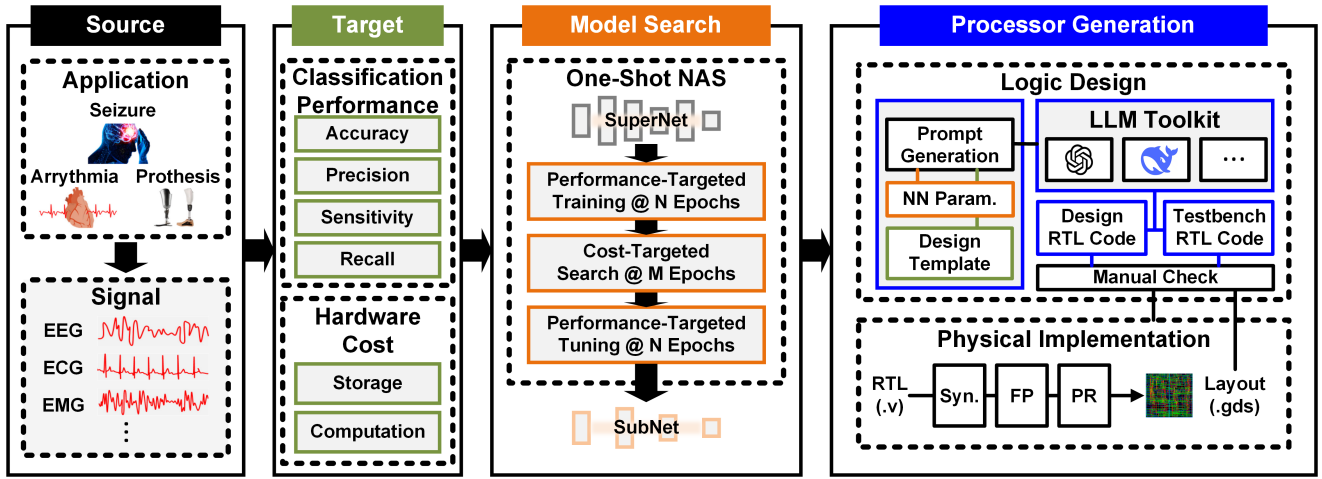


Fig. 2. The proposed end-to-end agile design generation framework of biosignal processors, termed BioSeek, with four key steps: Source, Target, Model Search, and LLM-based Processor Generation.

resource cost perspective, on-chip integration of DNN processors can lead to increased resource overheads and complexity, impacting both the silicon area and power consumption. Balancing the resource allocation to meet the computational demands of DNNs while optimizing area efficiency and power consumption presents a trade-off that designers must carefully navigate to achieve a cost-effective and high-performance solution.

To meet the growing demand for advanced and efficient healthcare solutions at the edge, we introduce BioSeek, an agile design generation framework empowered by cutting-edge large-language models (LLM). BioSeek aims to address the design complexities of biosignal processors optimally and efficiently. The subsequent sections of this paper are structured as follows: Section II outlines the structure of the BioSeek framework, Section III delves into the software and hardware implementation approaches, Section IV showcases the implementation details and experimental outcomes, and Section V provides the concluding remarks for this study.

II. BIOSEEK FRAMEWORK

The proposed BioSeek framework for the design generation of biosignal processors is a structured approach that integrates advanced neural architecture search (NAS) techniques and large-language models (LLMs) to optimize the performance and efficiency for various edge healthcare applications. The framework is delineated into four key steps: Source, Target, Model Search, and Processor Generation. Below is a detailed description of each step as depicted in Figure 2.

A. Source

The Source step identifies the applications and associated biosignals that the framework aims to address. From these applications, key biosignals such as EEG, ECG, and EMG are extracted, as they play a crucial role in diagnosing and managing various health conditions. Additionally, this step clarifies whether the design will focus on a specific application

or accommodate multiple scenarios, which will inform the subsequent design decisions.

B. Target

In the Target step, the framework defines the desired performance metrics and hardware constraints that the biosignal processor is expected to meet. Classification performance metrics such as accuracy, precision, sensitivity, and recall are established to ensure that the processor can effectively classify and respond to the biosignals. Hardware cost considerations regarding storage and computation are considered to optimize both the software and hardware designs. This step aligns the design objectives with the operational requirements, ensuring that the generated processors will be effective and efficient.

C. Model Search

The Model Search step utilizes a one-shot Neural Architecture Search (NAS) methodology [13] [14] to efficiently explore potential neural network architectures tailored to the specified targets. Initially, the framework trains candidate models for a predetermined number of epochs to assess their performance based on classification metrics. Following this evaluation, the framework searches for architectures that minimize hardware costs while preserving performance. Once promising architectures are identified, fine-tuning is conducted to further enhance their performance. The result of this step is a selection of optimal architectures, termed "SubNet," which are prepared for the next phase of processor generation.

D. Processor Generation

The final step, Processor Generation, converts the selected neural architecture into hardware designs. This process consists of two main components: Logic Design and Physical Implementation. Using a large-language model (LLM) toolkit [15], the framework generates the Register Transfer Level (RTL) code for both the design and the testbench. This is achieved through prompt generation that incorporates neural

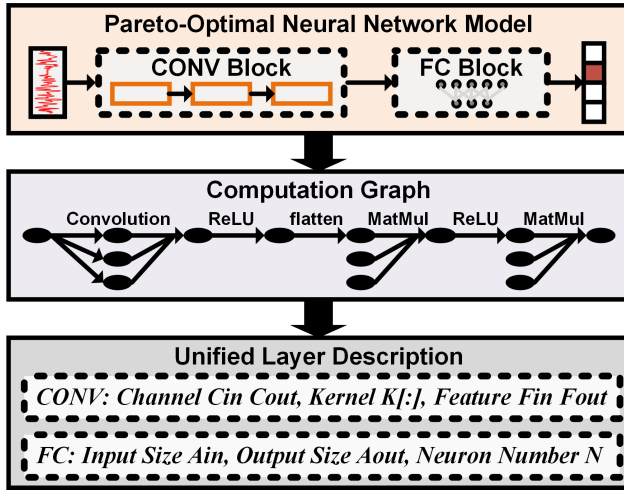


Fig. 3. Generation flow of unified text-based model descriptions from original codes.

network parameters and design templates, ensuring the generated code is specifically tailored to the chosen architecture. Following this, the RTL code is synthesized into a physical layout using a modern automatic physical design flow, which includes stages such as floorplanning (FP) and placement and routing (PR), ultimately resulting in the creation of the layout file (.gds). A manual verification step is conducted to ensure the integrity and accuracy of the generated design files both before and after the physical implementation.

III. SOFTWARE/HARDWARE IMPLEMENTATION

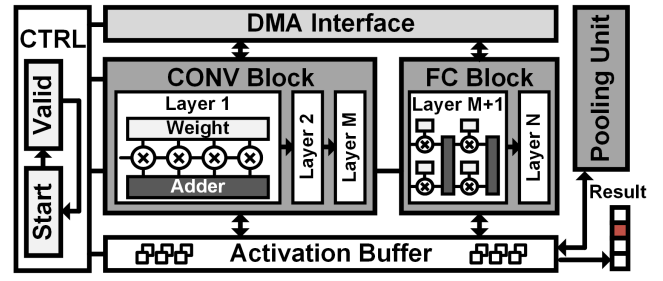
A processor has been designed to efficiently perform AER event pre-processing and SNN task evaluation in the proposed pipeline scheme.

A. Software Implementation

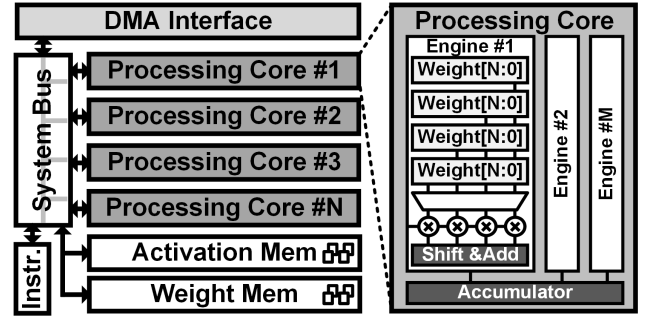
Figure 3 illustrates the process of generating unified text-based model descriptions from the original neural network code. The Pareto-optimal neural network model is derived from the one-shot NAS engine, ensuring optimal performance. For edge healthcare applications, the model typically consists of two primary components: convolutional (CONV) layers and fully-connected (FC) layers. Given that the data features of input biosignals are often characterized by temporal variance, the CONV block usually employs 1-D kernels, a detail reflected in the text-based model descriptions and corresponding hardware implementations.

Once the neural network model is defined, the next step is to construct the computation graph. This graph depicts the flow of data through the various layers of the neural network, capturing the relationships between different operations. It consists of multiple nodes and edges, with each node representing a specific operation, such as convolution, ReLU activation, flattening, or matrix multiplication.

The final stage of the generation process involves creating a text-based layer description that translates the computation



(a) Layer-Cascaded Near-Memory Architecture



(b) Layer-Wise In-Memory Architecture

Fig. 4. Proposed two memory-centric biosignal processor templates: layer-cascaded near-memory architecture and layer-wise in-memory architecture.

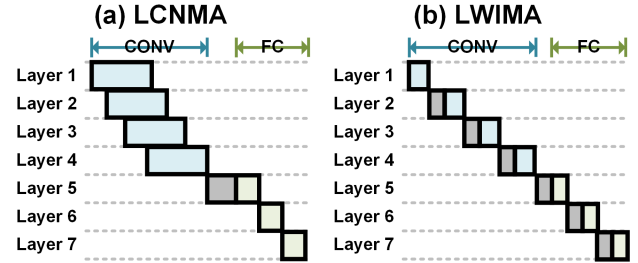


Fig. 5. Example timing diagrams of LCNMA and KWIMA for comparison.

graph into a standardized format. This description encapsulates the essential parameters and configurations of each layer in a clear and concise manner. Serving as a bridge between the original neural network code and the design specifications needed for biosignal processor implementation, this standardized layer description facilitates communication and integration with large-language models.

B. Hardware Implementation

In this work, we propose two hardware architecture templates for the generation of biosignal processors, as illustrated in Figure 4. We consider two deployment scenarios: one where the processor is tailored for one specific biomedical application and another where it serves as a reconfigurable engine capable of handling multiple applications. For application-specific biosignal processors, we introduce the Layer-Cascaded Near-

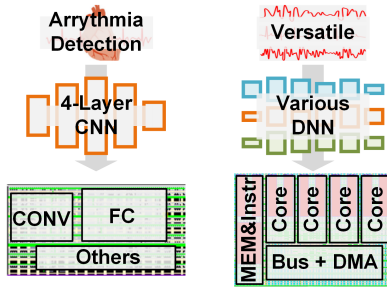


Fig. 6. Two design cases targeted for EAD and VBP generated via BioSeek.

Memory Architecture (LCNMA), while for versatile biosignal processors, we present the Layer-Wise In-Memory Architecture (LWIMA).

The LCNMA is designed to enable low-latency data flow and computation by positioning processing units close to memory resources. This architecture comprises several components, including a DMA Interface, a CONV Block, an Activation Buffer, an FC Block, and a Pooling Unit. The CONV Block consists of multiple layers (from Layer 1 to Layer M), each configurable with its corresponding weights, which are stored in dedicated buffers for quick access during computations. The FC Block connects all neurons from the previous layer to the next in a crossbar structure, facilitating highly parallel and low-latency execution.

In contrast, the LWIMA adopts a more integrated approach by tightly coupling processing and memory components. It includes a DMA Interface, Processing Cores, Activation Memory, Weight Memory, and instruction decoders connected via a system bus. The parallel in-memory processing cores are designed to execute layer operations concurrently, enhancing throughput and reducing processing time. Each processing core contains multiple engines, which are specialized for performing arithmetic operations like multiplication and accumulation. During execution, weights are pre-loaded into the memory cells of the parallel engines for reuse, while activations flow across the cores through the system bus. The execution of different layers relies on fetched instructions, requiring manual intervention to manage the design of this aspect.

Figure 5 presents the timing diagrams for both the Layer-Cascaded Near-Memory Architecture (LCNMA) and the Layer-Wise In-Memory Architecture (LWIMA). LCNMA is designed to optimize execution through a sequential processing approach. In this architecture, the execution of each layer occurs in a strictly cascaded manner, with CONV layer operations performed in a layer-fused manner to further minimize overall latency and reduce total buffer size. In contrast, LWIMA adopts a more parallelized approach. Unlike LCNMA, LWIMA facilitates parallel processing within an in-memory structure, significantly enhancing both parallelism and scalability. In summary, while LCNMA offers a straightforward, low-latency solution through its sequential execution,

TABLE I
COMPARISON WITH STATE-OF-THE-ART WORKS

Type	ECG-based Arrhythmia Detection			Versatile Biosignal Processing		
	TBioCAS 2020 [4]	TCAS-I 2023 [5]	This Work	TBioCAS 2023 [6]	TBioCAS 2023 [7]	This Work
Process (nm)	180	180	40	65	55	40
Area (mm ²)	0.9246	10.00	0.18	1.74	5.0625	1.008
Frequency	10KHz	-	500KHz	0.5-2.5MHz	20MHz	20MHz
Model Type	MLP	Spike-MLP	CNN	CNN	CNN, SNN, BNN	CNN
Accuracy (%)	99.2	90.5	99.42	99.16@ECG, 85.13@EMG, 99.84@EEG	99.38@ECG, 86.39@EMG, 99.94@EEG	99.42@ECG, 88.74@EMG, 99.43@EEG
Power* (Stand-By)	-	82nW	9.27uW	14.3uW	-	39.87uW
Latency	629.8ms	-	1.89ms	41ms@ECG, 154ms@EMG, 56ms@EEG	7.3ms@ECG, 9.4ms@EMG, 13ms@EEG	9.4ms@ECG, 11.2ms@EMG, 26.6ms@EEG
FoM**	6.77uJ	-	3.24uJ	2.25uJ@ECG, 5.25uJ@EMG, 2.06uJ@EEG	0.99uJ@ECG, 1.31uJ@EMG, 1.73uJ@EEG	0.37uJ@ECG, 1.09uJ@EMG, 0.86uJ@EEG

* The power in this table is reported in stand-by mode.

** FoM is calculated by energy consumption per task.

LWIMA provides advantages in area efficiency and scalability by enabling concurrent processing.

IV. IMPLEMENTATION AND EXPERIMENTAL RESULTS

To evaluate the effectiveness of BioSeek, we conducted tests using two design cases: ECG-based arrhythmia detection (EAD) and versatile biosignal processing (VBP). As illustrated in Figure 6, the processor generated for EAD features fixed CONV and FC blocks, while the processor for VBP includes four reconfigurable parallel processing cores. Both designs are implemented using a standard 40nm CMOS process, and post-layout simulations are performed to assess key performance metrics such as standby power, processing latency, and energy consumption.

For the EAD evaluation, we utilized the MIT-BIH dataset, while the VBP evaluation incorporated not only the MIT-BIH ECG dataset but also the CHB-MIT EEG dataset and the NinaProDB1 EMG dataset for model training and testing. A summary comparing the performance of the processors generated by BioSeek with existing state-of-the-art solutions is presented in Table I.

V. CONCLUSION

This paper presents BioSeek, an innovative design generation framework that leverages advanced large-language models to address the complexities of developing biosignal processors for healthcare applications. By effectively balancing the demands of performance, efficiency, and cost, BioSeek facilitates the creation of both application-specific and versatile biosignal processors. The evaluation results demonstrate that processors generated through BioSeek achieve competitive performance compared to existing solutions, highlighting the framework's capability to meet the growing requirements for DNN-based healthcare solutions at the edge. As the healthcare landscape continues to evolve, BioSeek represents a significant step forward in enabling efficient and effective biosignal processing, ultimately contributing to improved patient outcomes and more responsive healthcare systems.

VI. ACKNOWLEDGEMENT

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